INS Lab Exp-4

Aim:

Implementation of Advanced Encryption Standard (AES) algorithm for Symmetric key encryption.

Description:

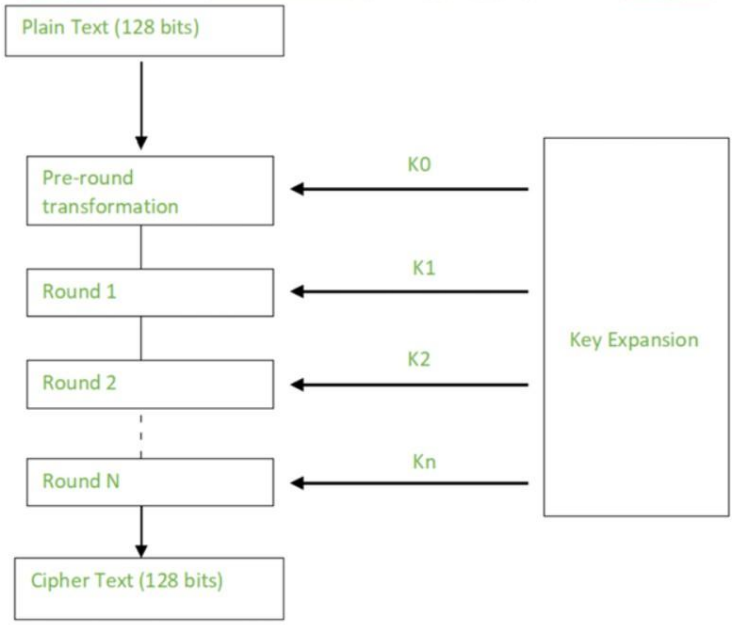
Advanced Encryption Standard (AES) is a symmetric-key encryption technique used to secure information. In symmetric encryption, the **same secret key** is used for both encryption and decryption.

AES works on data blocks of **128 bits**. It can operate with three different key lengths:

- **128-bit key** – 10 encryption rounds
- **192-bit key** – 12 encryption rounds
- **256-bit key** – 14 encryption rounds

AES follows a **substitution and permutation based structure**. During encryption, the input data goes through several transformations that change and rearrange the data to produce ciphertext.

The encryption key is expanded to generate separate keys that are used during the different rounds. The diagram in the reference record similarly shows the plaintext entering the initial transformation and then passing through successive rounds using generated round keys



Operations Performed in AES

Each normal AES round consists mainly of the following operations:

1. SubBytes

Each byte of the data is replaced with another byte using a predefined **S-Box (Substitution Box)**. This provides the substitution part of AES.

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2. ShiftRows

The bytes in the rows of the AES state are shifted by different positions. This rearranges the data and increases diffusion.

3. MixColumns

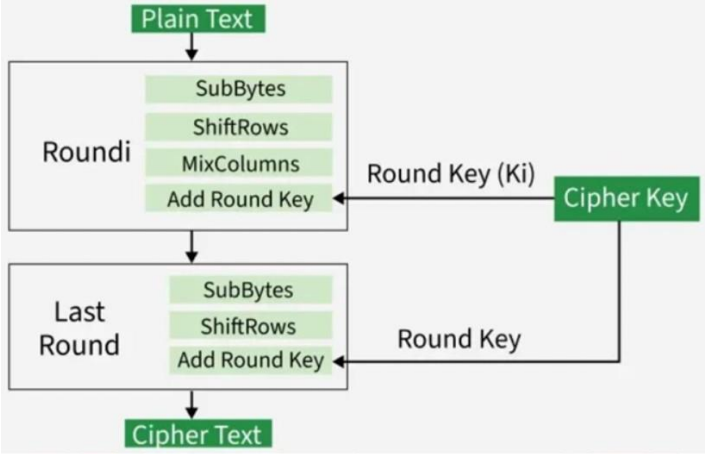
The bytes within each column are mathematically combined. This spreads the effect of individual input bytes throughout the block.

This operation is **not performed during the final round**.

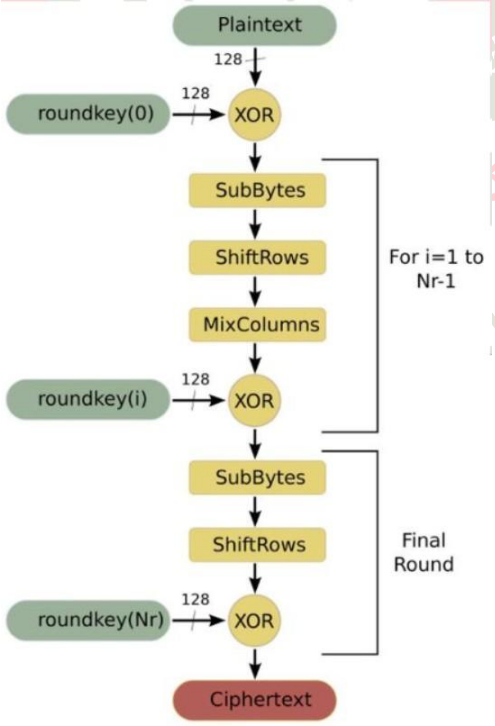
4. AddRoundKey

The current data is combined with the corresponding round key using the **XOR operation**.

These four operations and the fact that MixColumns is omitted from the last round are also specified in the reference record.



Flowchart:



Implementation:

```
from Crypto.Cipher import AES
from Crypto.Util.Padding import pad, unpad
key=b"1234567890123456"
cipher=AES.new(key, AES.MODE_ECB)
plaintext=input("Enter text: ")
encrypted=cipher.encrypt(pad(plaintext.encode(), AES.block_size))
print("Encrypted text:", encrypted.hex())
decipher=AES.new(key, AES.MODE_ECB)
decrypted=unpad(decipher.decrypt(encrypted), AES.block_size)
print("Decrpted Text:", decrypted.decode())
```

Output:

```
Enter text: helloworld
Encrypted text: 3d8d68734e34909dcbf9841aff8dbeea
Decrpted Text: helloworld
```

Conclusion:

AES (Advanced Encryption Standard) is a highly secure and efficient symmetric block cipher used to protect data confidentiality. It supports 128, 192, and 256-bit keys, providing strong protection against brute-force attacks. AES is also very fast on both hardware and software, making it suitable for high-performance applications. It is a globally recognized standard approved by NIST under FIPS 197 and is widely used in technologies such as Wi-Fi (WPA2/WPA3), SSL/TLS, file encryption, and disk encryption. However, since AES is a symmetric encryption method, the sender and receiver must securely share the same secret key, making key management and distribution an important consideration. Overall, AES offers an excellent balance of security, speed, efficiency, and flexibility.

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